

⑤ Polynomial Regression

Simple Linear
Regression

$$y = \beta_0 + \beta_1 x$$

Multiple Linear
Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_n x_n$$

Simple Polynomial
Regression

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots$$

For smaller values of degree this method tends to underfit
For larger values the model tends to overfit